

What is Firebase Storage used for?

Firebase Storage is a cloud service that stores and serves user-generated content such as:

- Image → Audio
- Videos → Documents

It provides secure, scalable and serverless file storage for apps.

~~How is~~

Firebase Storage

- Store Files
- Binary Files
- For media, documents
- Retrieved by download URL
- Folder based structure

Firestore

- Store structured data
- JSON-Like document
- for user info, messages
- Direct access to data
- collections & documents

Firestore = Spreadsheet (organized data)

Storage = Folder System (file & subfolders)

What is Storage Reference

A storage reference created using `ref()` is a pointer to a file's path in firebase storage.

Example:

```
const fileRef = ref(storage, 'images/user123/profile.jpg')
```

It can be used to:

- Upload file to a location
- Get download URL
- Delete a file
- Retrieve a file

★ It's like a path of some file on computer.

Why convert an image to Blob before uploading?

Firebase Storage needs binary data (Blob), while React Native's Image Picker gives URI (file path)

- URI → Just address of file
- Blob → the actual file data

Conversion example:

```
const response = await fetch(imageUri);  
const blob = await response.blob();
```

Then upload to Firebase

How to get file's Download URL?

After upload:

```
await uploadBytes(storageRef, blob);  
const url = await getDownloadURL(storageRef);
```

For existing file:

```
const url = await getDownloadURL(ref(storage,  
    'image/photo.jpg'));
```

The URL can be used for:

- Display an image in app
- Share files
- Save in firestore for access

Can we upload videos & pdfs in Firestore storage?

Yes, it support all type of files.

Images: .jpg .png .webp

Videos: .mp4 .mov

Documents: .pdf .docx

Archives: .zip .rar

How can we restrict upload access using firebase Rules?

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Firebase Storage Rules define who can read/write files.

1. Open Access (for testing)

allow read, write: if true;

2. Authenticated Users only

allow read, write: if request.auth != null;

3. User-Specific Folders

allow read, write: if request.auth.uid == userId;

4. File Size limit

allow write: if request.resource.size < 5 * 1024 * 1024

5. File Type Restriction

allow write: if request.resource.contentType.
matches ('image/*');

Update rules in Firebase console

Storage → Rules → Publish

How to delete a file from Firebase Storage?

Use deleteObject() function with file reference.

```
const fileRef = ref(storage, 'image/photo.jpg');  
await deleteObject(fileRef);
```

Optionally also remove URL from firebase ~~Storage~~ ^{Firestore}

```
await updateDoc(doc(db, 'users', userId),  
  {profilePicture: null});
```

What are best practices for naming uploaded files?

- Use timestamps for uniqueness
- Add user IDs
- include file extension
- Organize by folder
- use lowercase with nospace

```
ref(storage, 'profile/${userId}-${Date.now()}  
  .jpg');
```

Why store download URL in firestore?

- Getting from Firebase avoid extra API calls
- Keep file URLs linked with user data
- Easy fetch user query who uploaded files
- Cached, Lightweight, Support multiple files